

Identity and social exclusion: an experiment with Hispanic immigrants in the U.S.

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Received: 11 September 2014 / Revised: 21 July 2016 / Accepted: 27 July 2016 /
Published online: 9 August 2016
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Abstract In this study we conduct a lab-in-the-field experiment with low-income Hispanics in three neighborhoods in a large city in the U.S. to investigate how identity and social exclusion influences individual contributions to fund local public goods. We find that while the strength of identity has a significant and positive impact on individual contributions to local public goods, the perception of social exclusion significantly decreases contributions. Our findings thus suggest factors that may impede full civic participation, and shed important light on potential policies to increase integration of immigrants and ethnic minorities into mainstream society.

Keywords Identity · Social exclusion · Public goods · Immigration

JEL Classification H4 · C91 · C93

Electronic supplementary material The online version of this article (doi:[10.1007/s10683-016-9492-1](https://doi.org/10.1007/s10683-016-9492-1)) contains supplementary material, which is available to authorized users.

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1 Motivation and introduction

As the world becomes increasingly integrated, many communities have become more diverse, facing a rapid growth of ethnic minority groups and an influx of immigrants. On the one hand, diversity boosts innovation, stimulates local economic growth, and leads to culturally more vibrant communities. On the other hand, many communities experiencing increasing diversity face serious challenges, for example, in supporting a high-quality public sector (Alesina et al. 1999; Habyarimana et al. 2007). Two factors—identity and social exclusion—play major roles in the social integration process of ethnic minority groups and immigrants into their new communities (Byrne 2005). This paper demonstrates how these factors influence voluntary contributions toward public goods.

Identity and social exclusion are widely considered in social science disciplines such as sociology, politics, anthropology, psychology and economics. Identity is a person's sense of self derived from perceived membership in social groups, e.g., race, gender, or religion (Tajfel et al. 1971; Tajfel and Turner 1979, 1986). Social exclusion refers to disadvantages that individuals or groups experience when systematically blocked from accessing opportunities and resources which are normally available to members of the mainstream society (Stewart et al. 2006). While each of these factors can impede the social assimilation of an individual into a new society, they are conceptually distinct.

Take the Hispanic sample in our study as an example. Some participants were born in the U.S., and thus were very likely to have high identity and a low degree of social exclusion. But it is possible for some of them, despite the low degree of social exclusion, to have low identity, especially if they live in an ethnically heterogeneous community or a community dominated by another ethnic group. On the other hand, in a traditional Hispanic enclave, it is possible that Hispanic immigrants may have a high identity but a strong feeling of being excluded from the mainstream American society. We believe that the constructs of social identity and exclusion are not simply mirror opposites, and thus that the empirical relationship between them is of interest. Our dataset is unique in capturing these dimensions independently, and our results indeed indicate that they are not well-correlated, at least among this sample.

Social assimilation of a person into a new society is a multi-dimensional process that can be impeded by higher levels of social exclusion and lower levels of identification. Despite their importance in explaining such phenomena as ethnic conflicts, racial conflicts and discrimination, there are few studies in economics that address the impact of identity and social exclusion in decision-making.

One exception is Akerlof and Kranton (2003). They provide a theoretical framework to capture the impact of identity and social exclusion on economic decisions by a dominant group and a non-dominant group in a society. Their model and its solution not only show a major effect of identity, but also demonstrate how the non-dominant group's feeling of being socially excluded and their choices to seek psychological fulfillment in opposing the mainstream culture may also influence economic outcomes. For non-dominant groups of a society, Akerlof and Kranton (2003) suggest that identity and social exclusion are two crucial

components of the social integration process which profoundly affect engagement in economic activities considered to be the norms of mainstream society.

In this paper we experimentally investigate the impact of identity and social exclusion on economic decisions. We focus on one important dimension of economic decision making—individual's contribution to support public goods, which is a norm of mainstream society in many cultures (e.g., Ostrom 2000). Specifically, we conduct a lab-in-the-field experiment with low-income Hispanic residents in three neighborhoods in a large city in the U.S. to investigate the impact of identity and social exclusion on contributions to public goods.

In the experiment participants are given opportunities to contribute to a linear lab public good and to three organizations which provide local public goods. We find that identity and social exclusion, elicited through a post-experimental questionnaire, are both important determinants of contributions to public goods. While contributions to the lab public good and local organizations increase with the strength of identity, contributions to the local organizations decrease with perceptions of social exclusion.

As discussed in the literature review, much of the previous economics literature has manipulated identity (via priming). This is one of the few papers which elicit natural identities, and the only one examining their impact on public goods provision. This methodology thus complements the previous work by showing the importance of identity, both manipulated and measured, on economic decisions. In addition, to the best of our knowledge, this is the first study to experimentally investigate the impact of social exclusion on economic choices of any type.

We focus on a unique subject pool—Hispanics—one of the largest and fastest-growing ethnic minority groups in the U.S. Hispanics constitute 17 % of the nation's total population. This proportion is projected to be 31 % by 2060 (U.S. Census 2010). However, despite their growing importance in the economy and society, the racial phenotypes and strong accents make it difficult for many Hispanics to “pass” as an American. Massey and Sánchez (2009) show that 60 % of Hispanic immigrants reject the identity of being an American, while 82 % report a Latino identity.

The unique dataset we collect from this population provides naturally-occurring variation in individual-level measures of identity and social exclusion. Since the rapid growth of the Hispanic population is a primary driving force behind the increase in ethnic heterogeneity in American communities, our dataset allows us to study closely the possible challenges faced by these communities in public goods provision, with an eye to actionable policy interventions.¹

¹ As part of a large-scale field project, this paper differs from its four companion studies including Leonard et al. (2010), de Oliveira et al. (2011, 2012), Li et al. (2015). This paper is the only one in the project that focuses on social exclusion and the Hispanic sample, and one of two papers (besides Li et al. 2015) which investigate identity. It is different from Li et al. (2015) in three aspects: (a) the two datasets overlap very little; only 26 observations (13 % of the data) in this study are also reported in the non-identity-prime treatment in Li et al. (2015); (b) this paper uses survey measures of identity, rather than an identity prime; and (c) this paper focuses also on social exclusion. Table S1 in the Supplementary describes these four papers including their main research questions, methodologies, findings and the data used.

The remainder of the paper is organized as follows. Section 2 reviews the related literature and introduces hypotheses. Section 3 describes the experimental design. The empirical analysis and results are presented in Sect. 4. Section 5 concludes and discusses the policy implications.

2 Previous literature, research questions and hypotheses

The main research questions in this study are how an individual's identity and social exclusion affect her contributions to lab public good and organizations which provide local public goods. In laboratory public goods experiments, participants contribute independently to a group account and the total amount contributed is evenly split among all the participants. In studies on charitable organizations, participants decide how much to contribute to a charitable organization, and the amount contributed goes to the organization directly rather than being split by the participants. In this section, we summarize previous research that encompasses both laboratory public goods and charitable organizations to formulate our hypotheses.

As shown in a growing volume of economics research, monetary incentives and various non-monetary factors affect voluntary contributions to the provision of public goods and charitable organizations (see Ledyard 1995 for an excellent review of the early literature and Chaudhuri 2011 for a more recent review; more recent studies include List and Lucking-Reiley 2002; Karlan and List 2007). Some examples of non-monetary factors include warm glow (Andreoni 1989, 1990), self-image (Bénabou and Tirole 2006; Ariely et al. 2009), social preferences (Fischbacher and Gächter 2010), social information (Frey and Meier 2004; Shang and Croson 2009), and social pressure (DellaVigna et al. 2012). Although much of the literature shows that many motives of giving apply to both laboratory public goods and charitable organizations, we will make clear distinctions between these two categories in our discussion on the impact of identity and social exclusion below.

A robust finding in more than three-decade of research in psychology is that individuals show intergroup bias or ingroup favoritism when they are categorized into different social groups (Tajfel et al. 1971; Tajfel and Turner 1979, 1986; see Brewer 1991 for an excellent survey on this literature). The concept of identity is formally introduced into economic analysis in the seminal work by Akerlof and Kranton (2000). Since then, a growing volume of economics literature extends the psychology literature to the impact of identity on economic decision making. Studies find that group identity influences one's willingness to punish norm violation (Bernhard et al. 2008), and other-regarding preferences (Chen and Li 2009). In addition, interacting with ingroup members leads to higher level of cooperation in social dilemmas (Eckel and Grossman 2005; Goette et al. 2006; Charness et al. 2007), better coordination in a Battle of the Sexes game (Charness et al. 2007) and in the minimum-effort game (Chen and Chen 2011) when compared with outgroup members.

There is a growing experimental economics literature on the impact of naturally-occurring identity on a variety of economic decisions. Hoff and Pandey (2006), in

an experiment conducted in India, show that publicly announcing caste status negatively affects low-caste boys' performance in an incentivized maze-solving task. Benjamin et al. (2010) investigate the impact of racial and gender identity on subjects' choices in time and risk preference tasks. Afridi et al. (2015) find that the performance of low-status household registration holders in China is negatively affected in a cognitive task when their low-status identity is made salient. Li et al. (2015) find that priming an individual's community identity affects one's giving to local charitable organizations. In contrast with these studies which prime identity, our research adopts an alternative approach of eliciting an individual's strength of identity.

Our hypotheses on the impact of identity on contributions to laboratory public goods and local organizations could be formulated by applying Akerlof and Kranton's (2000, 2003) model to the public goods setting. In their model they propose a neoclassical utility function in which social norms of behaviors are prescribed by one's social identities, and deviations from these norms cause disutility. In many cultures it is the norm of mainstream society for individuals to contribute to support public goods and charities (Ostrom 2000). When an individual strongly identifies with her society, she may care more about the welfare of the beneficiaries of public goods and/or derive greater warm glow feelings from helping others, hence would suffer more disutility when deviating from the norm of contributing, compared with those who experience lower level of identity. Therefore, we hypothesize that individuals with higher level of identity will contribute more to lab public good as well as organizations that provide local public goods to their society, compared to those with lower level of identity. These discussions lead to Hypotheses 1a and 1b.

Hypothesis 1a (*Identity and Laboratory Public Goods*): When interacting with others from the same society, individuals with higher levels of identity with their society will contribute more to laboratory public goods (VCM) than will those with lower levels of identity.

Hypothesis 1b (*Identity and Local Organizations*): Individuals with higher levels of identity with their society will contribute more to organizations which provide local public goods to their society than will those with lower levels of identity.

The theoretical framework in Akerlof and Kranton (2003) also suggests that when socially excluded, the non-dominant group experiences utility loss when engaging in activities consistent with the social norm for the dominant group. In the setting of laboratory public goods (VCM), however, we do not expect that the perception of social exclusion that participants experience in their daily life will affect contributions, since by design everyone has equal and full access to the benefits of the public good funded by the total contributions in the group account, i.e., no one is being excluded from receiving his/her equal share of the total amount in the group account.

In contrast, in the setting of public goods provision by local organizations, social exclusion may decrease the motive to contribute in several ways. First, social exclusion creates unequal access to the benefits of public goods. Thus the actual private benefit one receives from the public good will depend on the degree of social exclusion. Compared with those who are fully integrated into the society, those who

are (or those who feel) excluded may receive (or perceive to receive) lower private benefits from public goods, and hence have a weaker motive to contribute. For example, Scotchmer (2010) constructs a theoretical model to show how lower levels of access to public goods reduce the benefits individuals obtain and hence their support. Second, socially excluded individuals or groups may not have the same incentive to help other members of the society (lower altruism or warm glow), which would also reduce contribution. This leads to Hypothesis 2a and 2b.

Hypothesis 2a (*Social Exclusion and Laboratory Public Goods*): Individual's contribution to laboratory public goods (VCM) is not affected by the (perceived) degree of social exclusion that the participants experience in their daily life.

Hypothesis 2b (*Social Exclusion and Local Organizations*): Individual's contribution to organizations that provide local public goods is negatively affected by the (perceived) degree of social exclusion that the participants experience in their daily life.

3 Experimental design

We design a lab-in-the-field (Morton and Williams 2010, p. 296) or extra-laboratory experiment in the field (Charness et al. 2013) to investigate how identity and perceived social exclusion influences individual decisions in contributing to public goods.

3.1 Participants

Our unique sample of low-income Hispanics was drawn from three neighborhoods in Dallas, Texas. Neighborhood 1, with 13.9 % Hispanics, is a low-income urban neighborhood which has long been home to a largely African-American community. Neighborhood 2, a low-mid-income urban neighborhood with 66.9 % Hispanics, has traditionally been a Hispanic enclave. Neighborhood 3 with 37.2 % Hispanics is a suburban area with diverse ethnic representation. The ethnic fractionalization index in each neighborhood is 0.26, 0.69, and 0.46, the median household income in each neighborhood is \$23,433, \$24,884, and \$54,518 (U.S. Census 2010).² We will use the fractionalization index and the neighborhood median income as neighborhood-specific controls in our analysis.

Our sample includes 201 Hispanics, with 20, 67, and 13 % coming from the three neighborhoods, respectively. Sixty-two percent of these participants are women, with an average of 1.7 children. Their average level of education is between elementary school and some years of high school. Their average household income is between \$10,000 and 20,000 per year. They have lived in their neighborhoods for an average of 16 years, and 13 % speak only English at home. Sixty participants

² We follow Alesina et al. (1999) in computing the ethnic fractionalization index using the U.S. 2010 census data for each neighborhood. The index for neighborhood n equals $1 - \sum_{g=1}^G s_{gn}^2$ where s_{gn} is the population share of ethnic group g in neighborhood n . The index measures the probability that two individuals randomly chosen from the neighborhood belong to different ethnic groups. The index varies from zero (homogeneous) to one (extremely heterogeneous).

Table 1 Participants' demographics

	Mean	Standard deviation
Female	0.621	0.486
Number of children	1.721	1.376
Education ^a	2.527	1.084
Household income ^b	2.049	1.038
Only English spoken at home	0.129	0.336
U.S. born ^c	0.299	0.459
Years living in neighborhood	16.01	15.64
Gambling choice (risk) ^d	2.099	1.262
Number of patient choices	2.323	2.965

^a Education is coded as 1 for “some elementary school”, 2 “finished elementary school”, 3 “some high school”, and 4 “finished high school or higher level of education”

^b Household income is coded as 1 for “less than 10 k”, 2 “10–20 k”, 3 “20–30 k”, 4 “30–40 k”, and 5 “greater than 40 k”

^c U.S. born refers to those participants who self-reported as being born in the U.S

^d In the gambling task participants choose one among six 50/50 gambles to play. The selected gambles are coded as 1 for “\$80/80”, 2 “\$60/120”, 3 “\$40/160”, 4 “\$20/200”, 5 “\$0/240”, and 6 “–\$20/260”

(30 %) self-reported as being U.S.-born and hence are U.S. citizens at birth.³ Table 1 reports the demographics of our sample. We will use these measures as individual-specific controls in our analysis.

3.2 Incentivized contribution tasks

In the experiment, each participant worked through an activity booklet containing a series of incentivized tasks: a simple gamble-choice task (Eckel and Grossman 2008), ten time preference tasks (Eckel and Grossman 2005), a laboratory public goods game (lab VCM, Ledyard 1995), and three real donation games with organizations that provide local public goods.⁴ We focus on the lab VCM and the three real donation games in this paper, and control for individual choices in the risk and time preference tasks in the analysis. Table 1 (above) shows the average responses in the time and risk tasks.

The experimental instructions and forms were designed for a low-literacy population, primarily pictorial with a minimum of text (see Supplementary Fig. S1, S2). Instructions were given verbally from a written script in Spanish. In the lab VCM, participants were randomly and anonymously assigned into groups of three.

³ Some non-U.S.-born participants in our sample could be U.S. citizens through naturalization but we are unable to identify them. We purposefully avoided immigration-related questions that could potentially drive away undocumented immigrants.

⁴ In the risk task, participants choose which of the six 50/50 gambles (\$80/80, \$60/120, \$40/160, \$20/200, \$0/240, and –\$20/260) to play. The ten time-preference choices were \$64 tomorrow versus \$68 in one (or five) month; \$60 tomorrow versus \$61 (62, 64, 66, 68) in 1 month; \$60 tomorrow versus \$65 (70, 80, 90, 100) in 5 months.

Table 2 Local organizations

Neighborhood	Service category		
	Health	Education	Business assistance/job training
1	Martin Luther King, Jr. Family Clinic	Dallas Bethlehem Center	Inner City Community Development Corporation
2	Barrios Unidos Community Clinic	Voice of Hope Ministries	The Plan Fund
3	Collin County Adult Clinic	Head Start	Practical Parenting Education

They were each given an endowment of \$60, and decided how to allocate it between a private and a group account by choosing from four discrete options: put \$60, 40, 20 or 0 in the private account, and the rest in the group account.⁵ Money put in the group account was doubled, and divided equally among the three group members (Marginal Per Capita Return = 0.66).

The three real donation tasks were designed and framed similarly to the lab VCM, except that the total amount put into the group account was doubled and sent to a local organization. We consulted with local community leaders and selected three widely-known and well-respected organizations that provide local public goods. These three organizations provided health, education, or job training services, respectively, in each neighborhood (see Table 2 for a description of the organizations).

The design of the real donation games parallels the lab VCM except that each participant receives full benefit from the public good in the VCM, while access to local organizations may vary by the degree of social exclusion. This design artificially creates differences in one's access to two types of public goods, i.e., full access to the lab VCM in contrast with potentially restricted access to local charitable services.

To avoid confusion, all participants did the VCM task followed by the real donation tasks since the design of the latter was directly based on the former. The order of the three real donation tasks was counterbalanced across sessions, and no order effects were found.

3.3 Identity and social exclusion measures

A post-experimental survey was conducted to collect individual identity and social exclusion measures, and other socio-demographic information. The exact wording of the identity and social exclusion questions is provided in Supplementary material.

We collected two sets of identity measures. The first set relates to individual's identities as American, Texan, and member of their neighborhood. Participants rate

⁵ The four discrete options were designed to simplify the task, and allowed the choices to be depicted visually using \$20 bills. We eliminated the 50/50 option in order to avoid the 1/n split cognitive bias in which individuals divide assets evenly across the n options available, a naive type of diversification observed in individual investment decisions (Benartzi and Thaler 2001).

the strength of these three identities using a 4-point scale.⁶ The second set of identity measures focuses on one's community identity exclusively. The participants were asked to rate two statements on a 5-point scale about their sense of belonging and pride being part of their community. These two measures were included in a set of nine statements that also asked participants about their engagement in the community, interactions with other residents, and overall feeling living in their community. This set of questions is adapted from the Multigroup Ethnic Identity Measure (Phinney 1992) with a different focus on the characteristics of a community rather than those of an ethnic group.

We thus focus on five categories that *directly* measure one's identities—American, Texan, member of the neighborhood, sense of belonging, and pride in the neighborhood.⁷ Since our measures of identities are elicited rather than controlled, our methodology is complementary to previous work discussed in Sect. 2 which has manipulated identity. It allows us to study the importance of *measured* social identity, in addition to manipulated identity, on individual economic decisions.

For social exclusion, participants were given a list of financial, social and political activities and asked to choose those that they wanted to participate in but could not (Calonico et al. 2007), including having a bank account, saving for retirement, having a cooperative account among friends and coworkers, participating in political issues or organizations, voting, working, going to school, owning a home, having health insurance, and getting a loan. Some of these social exclusion measures are highly correlated with income and citizenship status, and hence are likely to reflect one's economic, financial constraints or legal status in the U.S. rather than *social* exclusion. In fact, some measures (e.g., voting, owning a home, and having health insurance) are highly correlated with the citizenship status. These issues will be addressed in our empirical analysis in Sect. 4 below.

3.4 Recruitment and implementation

The experiments were conducted from fall 2007 through summer 2008. Recruiting pamphlets were distributed and posted in public places. We also recruited participants via door to door visits in local communities and outside popular Hispanic supermarkets. Recruiting took place during multiple time slots throughout the day to accommodate different routines of residents. Interested participants called for an appointment to participate. Sessions were run between 9 AM and 4 PM during the week and on the weekends at three field stations (one in each neighborhood) in safe locations near multiple bus lines that were familiar to local residents.⁸ Each session lasted about 2 h. One decision was chosen at random for

⁶ We re-scale the 4-point scale measure to a 5-point scale measure so they can be combined for our average identity measure.

⁷ As shown in Supplementary material, the exact wording of the statements that the participants were asked to rate was "I see myself as an American", "I see myself as a Texan", "I see myself as a member of my neighborhood", "I have a strong sense of belonging to the (neighborhood name) area", and "I have a lot of pride in the (neighborhood name) area".

⁸ On average 83 percent of those who made an appointment showed up. No significant differences were found across the neighborhoods.

payment. The average earnings were \$60, about the average daily wage earnings of these participants.⁹

4 Analysis and results

In this section we investigate the impact of identity and social exclusion on contributions to the VCM and local public goods. We first present descriptive statistics, then discuss the regression analysis and results using the individual level data.

4.1 Descriptive statistics

Figure 1 shows the contribution histograms by categories. The average unconditional giving is \$27.00 (VCM), \$18.21 (Education), \$19.80 (Health), and \$17.21 (Job training). The likelihood of giving is 76.1, 65.2, 67.7, and 64.2 %, and the amount contributed conditional on giving is \$35.42, 27.94, 29.26, and 26.82, respectively.^{10,11}

Our main explanatory variables of interest are identity and social exclusion. For identity, we focus on the average of the five measures including American, Texan, member of the neighborhood, sense of belonging, and pride in the neighborhood as discussed in Sect. 3. Since these identity measures are highly and positively correlated with each other (correlations from 0.15 to 0.78, $p < 0.05$ for all), their average is used as each participant's overall strength of identity in the analysis. Figure 2 summarizes this average identity measure and each separate identity category. The higher the level, the greater the strength of identity. The correlations between this average and each identity category vary from 0.46 to 0.79 ($p < 0.01$ for all).

A summary of the participants' social exclusion measures is presented in Table 3. To measure social exclusion, we first note that respondents who were U.S. born may be less likely to be socially excluded than those who are not, simply

⁹ High stakes were used to attract participation of local residents in these neighborhoods. The experimental sessions were long (around 2 h), and recruitment of this particular population was difficult because of a distrust of authority. The relatively high stakes ensured that we had an enthusiastic participant pool.

¹⁰ Although Fig. 1 shows differences in giving between VCM and each local charity, we find no statistically significant differences in the amounts given, or the likelihood of giving, across these categories in the regression analysis presented in Sect. 4.2. How VCM giving compares to giving to local organizations is an interesting question, which is explored directly in de Oliveira et al. (2011). They use the same experiment design, but a sample of African-American residents in a low-income neighborhood, and find that individual choices in VCM predict giving to local organizations. However, choices vary systematically by the decision contexts and individual perceptions of their neighborhoods. In contrast, the main focus of this study is not on the comparison in giving between VCM and local organizations; it is the impact of social exclusion and identity on giving overall. We use the comparison between the VCM (where there is no social exclusion) and the local organizations as a way to identify the effects of these factors.

¹¹ Correlations between the contributions and risk/time preference choices are presented in Supplementary Table S2.

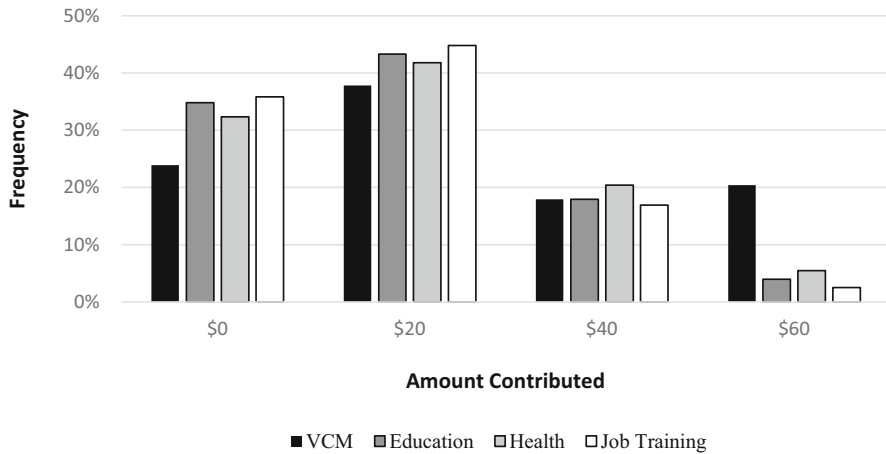


Fig. 1 Histogram of amount contributed to VCM and local organizations

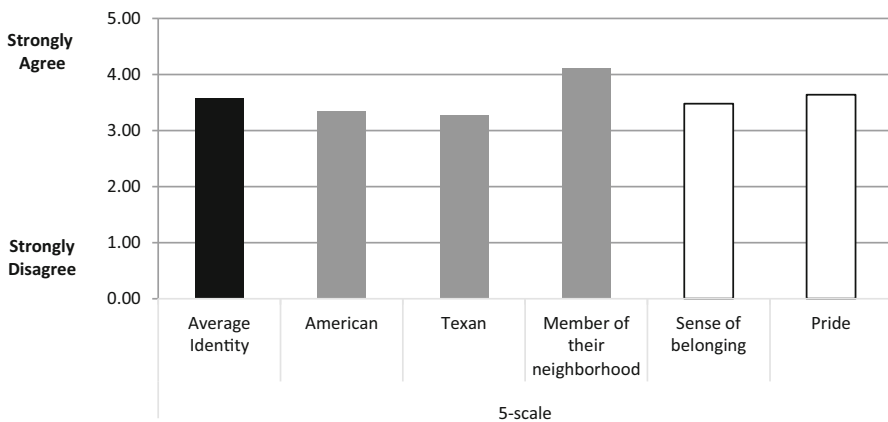


Fig. 2 Descriptive statistics on identity. The exact wording of these social identity questions is included in Supplementary material. American, Texan and Member of Neighborhood are each rescaled from the original 4-scale score to the 5-scale score to be comparable to Sense of Belonging and Pride (1 strongly disagree, 2 disagree, 3 neutral, 4 agree, 5 strongly agree)

because of what they are legally eligible to do. We thus categorize our participants' responses to our social exclusion measures based on whether they were U.S. born or not in Table 3 (subject to the caveat noted in footnote 3). We find three measures for which being U.S. born significantly influences responses: whether one is able to vote, own a house or have health insurance ($p < 0.01$). We eliminate these three measures and construct an overall social exclusion measure by averaging the responses of the other categories. We then use this average social exclusion measure in the analysis. A higher level captures a greater degree of social exclusion. The

Table 3 Descriptive statistics on social exclusion

	All (N = 201)	U.S.-born ^a (N = 60)	Non-U.S.-born (N = 141)	U.S.-born versus non-U.S.-born (P-value)
(a) Having a bank account	0.418 (0.494)	0.367 (0.486)	0.440 (0.498)	0.337
(b) Saving for retirement	0.552 (0.499)	0.517 (0.504)	0.567 (0.497)	0.508
(c) Having a cooperative account	0.269 (0.444)	0.283 (0.454)	0.262 (0.442)	0.759
(d) Participating in politics	0.214 (0.411)	0.183 (0.390)	0.227 (0.420)	0.490
(e) Voting	0.473 (0.500)	0.233 (0.427)	0.574 (0.496)	0.000
(f) Working	0.537 (0.500)	0.517 (0.504)	0.546 (0.500)	0.702
(g) Going to school	0.587 (0.494)	0.517 (0.504)	0.617 (0.488)	0.186
(h) Owning a home	0.542 (0.499)	0.433 (0.500)	0.589 (0.494)	0.043
(i) Having health insurance	0.597 (0.492)	0.383 (0.490)	0.688 (0.465)	0.000
(j) Getting a loan	0.284 (0.452)	0.283 (0.454)	0.284 (0.452)	0.996
Average of all categories ^b	0.447 (0.257)	0.372 (0.226)	0.479 (0.263)	0.006
Average excluding (e), (h) and (i)	0.409 (0.261)	0.381 (0.246)	0.420 (0.267)	0.327

^a U.S.-born refers to the 60 participants who self-reported as being born in the U.S. The remaining 141 participants were non-U.S.-born

^b The exact wording of these social exclusion questions is included in Supplementary material. For each category the response is coded '1' if that category is checked by the participant (i.e., being excluded), otherwise '0'. Standard deviations are in the parentheses. Test of proportion is used for each category, and t-test for the averages. Our analysis focuses on the average social exclusion measure (in the last row) that excludes the three categories (voting, owning a home and having health insurance) for which significant differences are found between those U.S.-born and non-U.S.-born

correlations between the average measure and each separate category vary from 0.42 to 0.64 ($p < 0.01$ for all).¹²

In our sample of Hispanic immigrants, the correlation between identity and social exclusion is low and statistically insignificant (0.07, $p = 0.31$).¹³ In addition, there is substantial variation in both variables. These data characteristics allow us to separately and independently identify the impact of identity and social exclusion in the empirical analysis.

¹² Instead of using the average measures of identity and social exclusion, an alternative approach is to generate composite measures for identity and social exclusion, respectively, using the confirmatory factor analysis (see Harrington (2008) for details). A brief description on this approach and the corresponding regression results (which are consistent with those in the main analysis presented in Sect. 4.2) are reported in Supplementary Table S3.

¹³ The lack of correlation between identity and social exclusion may be surprising at first glance. As we have argued above, however, these constructs are not simply mirror opposites. While some U.S.-born participants may have a strong identity and a low degree of social exclusion, others may have experienced low degree of social exclusion, yet have *low* identity, especially if they live in an ethnically heterogeneous community (e.g., neighborhood 3) or a community dominated by another ethnic group (e.g., neighborhood 1 of a largely African-American community). On the other hand, in a traditional Hispanic enclave (e.g., neighborhood 2), it is possible that Hispanic immigrants have a *strong* identity but still feel socially excluded from mainstream American society. Our dataset independently captures these two dimensions in order to empirically examine their relationship and independently identify their effects.

4.2 Regression analysis and results

We use the following specification to analyze the impact of identity and social exclusion on individual contributions.

$$C_{ij} = \beta_0 + \beta_1 Identity_i + \beta_2 Exclusion_i + \beta_3 Local + \beta_4 Identity_i \cdot Local + \beta_5 Exclusion_i \cdot Local + \Gamma + \varepsilon_{ij}.$$

Analysis and results are presented in Table 4. We focus on unconditional giving in columns 1, 4, and 7 using a Tobit model because the dependent variable is censored at \$0 and 60.¹⁴ We then use the Cragg Hurdle model to decompose the analysis into the likelihood of giving (columns 2, 5 and 8) and the amount contributed, conditional on giving (columns 3, 6 and 9). The Hurdle model, commonly used in health economics, captures the two parts of participants' decisions: (a) they need to decide whether to give at all, and (b) if they decide to give, they need to choose how much to give.¹⁵

Specifically, the dependent variable C_{ij} is the unconditional amount, the likelihood of giving, and the conditional amount given by individual i in game j (VCM, Education, Health or Job Training) in columns 1–3, respectively. The main independent variables include individual's strength of identity ($Identity_i$), perceived degree of social exclusion ($Exclusion_i$), the dummy variable for local organizations ($Local$), and the interaction terms between the local organizations with identity or social exclusion to allow its differential impact on giving to VCM and local organizations. The overall impact of identity and social exclusion on giving to local organizations ($\beta_1 + \beta_4$, $\beta_2 + \beta_5$) is reported in the middle panel of the table, followed by the estimates of other covariates discussed below.

Since our measures of identity and social exclusion are elicited rather than manipulated in the experiment, it is particularly important to control for neighborhood-specific and individual-specific characteristics in the analysis. Neighborhood characteristics include the ethnic fractionalization index and median household income (thousands). Individuals' socio-demographic characteristics include gender, education, whether they were U.S. born, language spoken at home, number of years living in the neighborhood, number of children, income, risk (measured by individual choices made in the gambling task) and time preference (the number of patient choices made in the 10 time preference tasks).¹⁶ As a robustness check, we include neighborhood-specific characteristics as the only controls in columns 4–6, and individual-specific characteristics as the only controls in columns 7–9. Standard errors are clustered on the individual level. Marginal

¹⁴ One referee suggested an ordered Probit model rather than a Tobit. The results reported below are consistent with either analysis technique, and results from the ordered Probit are available upon request.

¹⁵ The Cragg hurdle model ("churdle" and "margins" in Stata 14) is used to estimate columns 2–3, 5–6 and 8–9. The overall likelihood function for the Cragg Hurdle model is constructed as the product of two likelihoods: the likelihood of contributing a positive amount (probit) and the conditional likelihood of contributing a certain amount out of the \$60 endowment. See Woodridge (2002, pp. 536–538) and Botelho et al. (2009) for more details on the Hurdle model.

¹⁶ The risk variable is a categorical variable with the most risky choice being 6 and the least being 1 in the risk preference task.

Table 4 Impact of identity and social exclusion on giving

Dependent variable	(1) Unconditional (\$)	(2) Likelihood	(3) Conditional (\$)	(4) Unconditional (\$)	(5) Likelihood	(6) Conditional (\$)	(7) Unconditional (\$)	(8) Likelihood	(9) Conditional (\$)
Identity (β_1)	2.59** (1.052)	0.148*** (0.038)	3.776** (1.578)	2.095** (0.956)	0.134*** (0.039)	3.424** (1.609)	2.369** (1.039)	0.144*** (0.040)	3.631** (1.601)
Exclusion (β_2)	-1.563 (3.102)	-0.152 (0.111)	-2.381 (4.797)	-0.202 (3.096)	-0.106 (0.123)	-0.473 (5.276)	-0.351 (3.046)	-0.107 (0.117)	-0.724 (4.876)
Local (β_3)	-1.524 (3.743)	0.141 (0.132)	-4.918 (5.935)	-1.368 (3.442)	0.138 (0.129)	-5.200 (5.978)	-1.476 (3.632)	0.155 (0.136)	-4.521 (6.019)
Identity*Local (β_4)	-0.644 (0.986)	-0.061* (0.037)	-0.531 (1.576)	-0.623 (0.908)	-0.060* (0.036)	-0.521 (1.590)	-0.623 (0.958)	-0.067* (0.038)	-0.699 (1.596)
Exclusion*Local (β_5)	-2.419 (3.162)	-0.089 (0.121)	-3.484 (4.925)	-1.995 (2.879)	-0.084 (0.119)	-2.903 (4.842)	-2.392 (3.060)	-0.080 (0.123)	-3.183 (4.970)
Overall effects on local organizations									
Identity ($\beta_1 + \beta_4$)	1.915** (0.792)	0.087*** (0.032)	3.244*** (1.242)	1.472** (0.697)	0.075** (0.032)	2.903** (1.239)	1.746** (0.816)	0.078** (0.035)	2.932** (1.365)
Exclusion ($\beta_2 + \beta_5$)	-3.982** (2.030)	-0.241*** (0.091)	-5.865* (3.191)	-2.197 (2.100)	-0.190* (0.098)	-3.377 (3.793)	-2.743 (2.108)	-0.188** (0.0957)	-3.907 (3.474)
Other covariates									
Fractionalization	-29.597*** (6.022)	-1.550*** (0.312)	-51.173*** (11.048)	-24.748*** (5.600)	-1.551*** (0.343)	-49.099*** (12.180)			
Neighborhood median household income	0.366*** (0.079)	0.016*** (0.004)	0.631*** (0.157)	0.250*** (0.075)	0.014*** (0.005)	0.515*** (0.180)			
Female	1.951* (1.112)	0.090* (0.047)	2.710 (1.749)				1.703 (1.112)	0.078 (0.050)	2.256 (1.797)

Table 4 continued

Dependent variable	(1) Unconditional (\$)	(2) Likelihood	(3) Conditional (\$)	(4) Unconditional (\$)	(5) Likelihood	(6) Conditional (\$)	(7) Unconditional (\$)	(8) Likelihood	(9) Conditional (\$)
Education	-1.480*** (0.521)	-0.047** (0.022)	-2.433*** (0.851)				-1.442*** (0.503)	-0.057** (0.023)	-2.450*** (0.842)
U.S.-born	-0.198 (1.702)	-0.001 (0.070)	-0.906 (2.710)				-1.011 (1.659)	-0.032 (0.074)	-2.178 (2.738)
Only English spoken at home	-2.993 (1.890)	-0.130* (0.076)	-4.507 (2.950)				-2.654 (1.918)	-0.134* (0.081)	-4.098 (3.076)
Years living in community	-0.005 (0.051)	-0.002 (0.002)	0.009 (0.082)				-0.023 (0.051)	-0.003 (0.002)	-0.013 (0.085)
No. of children	-1.304*** (0.416)	-0.050*** (0.017)	-1.983*** (0.682)				-1.010*** (0.431)	-0.035* (0.019)	-1.587*** (0.739)
Income 10–20 K	3.818*** (1.256)	0.102* (0.054)	6.932*** (1.984)				3.500*** (1.249)	0.096* (0.057)	6.363*** (2.050)
Income 20–30 K	3.316*** (1.335)	0.087 (0.059)	5.562*** (2.157)				3.079*** (1.395)	0.083 (0.066)	5.187*** (2.374)
Income 30–40 K	3.690* (2.090)	0.136 (0.104)	6.813* (3.592)				3.931* (2.034)	0.172 (0.116)	7.477*** (3.670)
Income >40 K	1.607 (2.513)	0.083 (0.115)	4.236 (4.605)				0.541 (2.529)	0.046 (0.120)	2.232 (4.718)
Risk	0.784* (0.401)	0.055** (0.022)	1.290* (0.745)				0.603 (0.411)	0.047** (0.022)	1.011 (0.767)
No. of patient choices	0.591*** (0.163)	0.024*** (0.008)	0.962*** (0.293)				0.607*** (0.169)	0.028*** (0.009)	1.045*** (0.305)
N	804	804	804	804	804	804	804	804	804

Table 4 continued

Dependent variable	(1) Unconditional (\$)	(2) Likelihood	(3) Conditional (\$)	(4) Unconditional (\$)	(5) Likelihood	(6) Conditional (\$)	(7) Unconditional (\$)	(8) Likelihood	(9) Conditional (\$)
Subjects	201	201	201	201	201	201	201	201	201
Log likelihood	-2517.9	-2578.4		-2580.5	-2652.7		-2540.4	-2612.1	
Pseudo R ²	0.039	0.051		0.016	0.024		0.031	0.039	

Tobit models are used in columns 1, 4 and 7, and Cragg Hurdle models in columns 2–3, 5–6 and 8–9. Columns 1–3 include neighborhood-specific and individual-specific characteristics. For robustness check, columns 4–6 include only neighborhood-specific characteristics, and columns 7–9 only individual-specific characteristics. Marginal effects are reported. Standard errors clustered at the individual level are in parentheses

* $p < 10\%$, ** $p < 5\%$, *** $p < 1\%$

effects are reported. Our discussion below will focus on the analysis with the full set of control variables in columns 1–3.

As columns 1–3 in Table 4 show, identity significantly increases the amount of giving, the likelihood of giving and the conditional amount given to both VCM and local organizations. The marginal effects of identity on giving to VCM (β_1) are \$2.56 for the unconditional amount ($p = 0.015$), 14.8 % for the likelihood ($p < 0.001$), and \$3.78 for the conditional amount ($p = 0.017$). Its marginal effects on giving to the local organizations ($\beta_1 + \beta_4$) are \$1.92 for the unconditional amount ($p = 0.016$), 8.7 % for the likelihood ($p = 0.008$), and \$3.24 for the conditional amount ($p = 0.009$).

In contrast, we find, as predicted, that social exclusion does not significantly influence giving to VCM (β_2 , $p > 0.10$) because by design, no participants are excluded from receiving the private benefits from their VCM group account. Social exclusion, however, significantly decreases the amount contributed to the local organizations, with a marginal effect of $-\$3.98$ ($\beta_2 + \beta_5$, $p = 0.050$). This impact is primarily driven by the negative influence of social exclusion on the likelihood of giving—the marginal effect is -24.1 % ($p = 0.008$). Social exclusion also marginally reduces the conditional amount of giving to local organizations, with a marginal effect of $-\$5.87$ ($p = 0.066$).

Neighborhood characteristics also play important roles in influencing giving. The average effects of ethnic fractionalization on VCM and local organizations are $-\$29.60$ for unconditional giving, -155 % for the likelihood of giving, and $-\$51.17$ for conditional giving ($p < 0.001$ for all), respectively.¹⁷ This is consistent with the previous literature that increasing neighborhood fractionalization will decrease giving. Moreover, the median household income, as an important economic indicator of the neighborhood, has significantly positive impact on giving. Every \$1000 increase in neighborhood median household income is associated with a \$0.37 increase in the average amount of giving to VCM and local organizations, a 1.61 % increase in the likelihood of giving and a \$0.63 increase in the conditional amount ($p < 0.001$ for all).

At the individual level, participants' education, income, number of children, risk and time preferences are the most important among individuals' socio-demographic characteristics in influencing average contributions to the VCM and local organizations. Contributions decline with the level of education ($p < 0.05$).¹⁸ We find that the amount of contributions and the likelihood of giving generally increases with one's household income but not in a linear fashion. Our findings also show that the more children one has, the less likely and the smaller amount one contributes

¹⁷ As another robustness check, we conduct additional analysis on the possible interaction effects between fractionalization and identity (or social exclusion). We focus on contributions to local organizations only, and present the results in Supplementary Table S4. We find that the positive impact of identity on the likelihood of giving and the conditional amount is significantly lower in more ethnically fractionalized neighborhood ($p < 0.05$, columns 2–3 and 5–6). However, the interaction between fractionalization and social exclusion is not statistically significant, and excluding this interaction term does not impact the results of other variables.

¹⁸ This might be due to the fact that the majority of our participants had relatively low levels of education.

($p < 0.01$). Those who make more patient choices in the time preference tasks are significantly more likely to give and make significantly higher contributions ($p < 0.01$). This implies that patient individuals may view their contributions as investments in their neighborhood's future and thus give more generously. Those who choose a more risky option in the gambling task tend to give more, although only marginally, yet are significantly more likely to give ($p < 0.10$ for the unconditional and conditional amount, $p = 0.012$ for the likelihood).

Women contribute marginally more than men on average ($p < 0.10$ for the unconditional amount and the likelihood of giving). Language use is marginally significant only for the likelihood of giving ($p < 0.10$). The effects of being U.S. born and the number of years living in the neighborhood are not significantly different from zero.

Comparing columns 1–3 (that include a full set of control variables) with columns 4–6 (that include only the neighborhood characteristics as controls) and columns 7–9 (that include only the individual characteristics as controls), we find that the effects of identity on VCM and local organizations as well as the impact of social exclusion on VCM are robust. While the effects of social exclusion on giving to the local organizations ($\beta_2 + \beta_5$) are slightly smaller in size in columns 4–6 and 7–9, relative to columns 1–3, the results are directionally correct. Moreover, the negative impact of social exclusion on the likelihood of giving to local organizations remains marginally significant in column 5 (-19.0% , $p = 0.053$) and significant in column 8 (-18.8% , $p = 0.050$). Both estimates are economically sizable.

In sum, we find evidence in support of our hypotheses. The identity of participants in our study positively affects their contributions to the laboratory public goods and the local organizations. Their perceptions of social exclusion negatively affect their contributions to the local organizations, yet do not affect giving to the laboratory public goods, as predicted.

5 Conclusion

In this study we conduct a lab-in-the-field experiment in three U.S. neighborhoods to investigate the impact of identity and social exclusion on individual's contributions to public goods. We focus on Hispanics, one of the largest and fastest-growing ethnic minority groups in the U.S. We find while the strength of social identity influences contributions to the laboratory public goods and local organizations positively, the perception of social exclusion leads to a decrease in contributions to the local organizations.

This study enriches our understanding of socio-economic factors underlying public goods provision and a well-functioning public sector. Our results show that low level of identification and perception of social exclusion may impede pro-social behaviors and, more generally, full civic participation in the mainstream society by immigrants and ethnic minorities. Since ethnically heterogeneous communities tend to have more minority groups than homogeneous ones, our results suggest that policy interventions to reduce social exclusion and to foster a common identity

among residents in these communities can be an effective direction to explore. Policy makers and community leaders may spend resources to support events that are inclusive of ethnic minority groups to increase their identity with their communities. In addition, targeted policy interventions for ethnic minority groups can promote the spirit of commonly shared identity. These interventions could stress similarities among residents and their common needs, e.g., we all use the parks, all our children need to be educated, unemployment can happen to anyone.

Our study also suggests a few other fruitful extensions for future research. Firstly, the methodology designed in this study could be extended beyond Hispanic immigrants to other groups (e.g., non-immigrant populations, other ethnic minority or majority groups). Secondly, while this study shows that identity and social exclusion are important influences on contributions to public goods, an interesting but still unexplored question is what (observable) factors influence identity and social exclusion. While not the focus of our study, preliminary analysis shows that identity is significantly stronger for those who speak only English at home and who have lived in their communities for a longer period of time. The positive correlation between identity and speaking the dominant language is found in other studies (Han et al. 2010; Milani 2008; Schmidt 2000). However, future research is needed to identify causal relationships behind the correlations illustrated here.

In conclusion, we confirm the impact of identity on economic decision-making by demonstrating its influence on contributions to fund public goods using a different methodology (measured rather than manipulated identity). However, the most important new finding from this research is the crucial role that social exclusion plays in economic decision making. This is the first study to measure and identify the impact of social exclusion, and our unique dataset allows us to identify this impact separately from that of identity. This paper thus serves as a valuable first step into this previously underexplored research area.

Acknowledgments We thank Angela de Oliveira and Catherine Eckel for their input in the design and implementation of this study. We also thank seminar/conference participants at the 2011 North-American Economic Science Association Conference (Tucson), UT Dallas, the 2013 Celebrating 25 Years of Warm Glow Conference at UCSD, and the CUNY Queens College Seminar Series. Field research assistance was provided by Wayra Rodriguez. Funding was provided by the National Science Foundation (SES-0752855). Any errors remain our own.

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